

L19 - Breadth First Search/Traversal/Visit

Thursday, May 28, 2020 8:16 AM

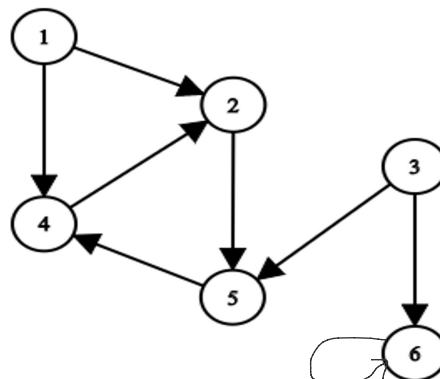
To traverse/reach each reachable vertex of a graph from a given starting vertex.

BFS(G,s)

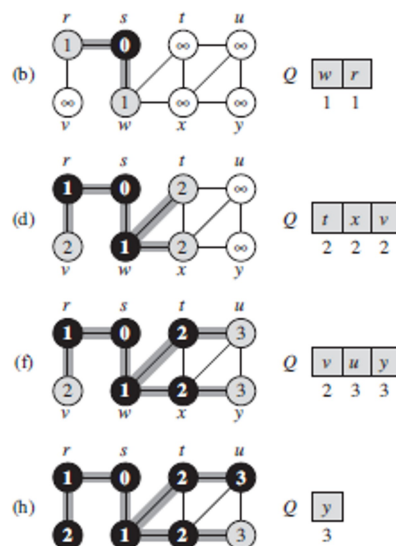
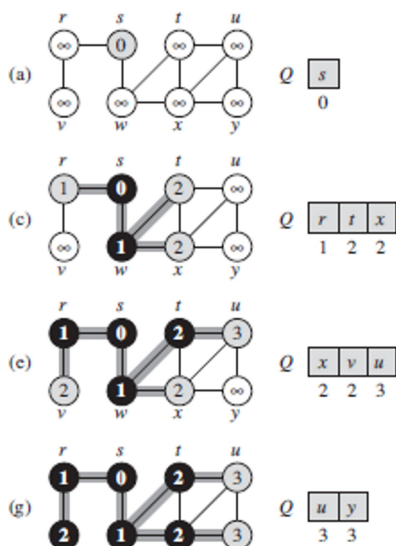
1. for each vertex $u \in G.V - \{s\}$
2. $u.color = WHITE$
3. $u.d = \infty$
4. $u.\pi = NIL$
5. $s.color = GRAY$
6. $s.d = 0$
7. $s.\pi = NIL$
8. $Q = \emptyset$
9. ENQUEUE(Q, s)
10. while Q
11. $q = DEQUEUE(Q)$
12. for each $v \in G.Adj[q]$ // explore edge (q, v)
13. if $v.color == WHITE$
14. $v.d = q.d + 1$
15. $v.color = GRAY$
16. $v.\pi = q$
17. ENQUEUE(Q, v)
18. $q.color = BLACK$ // finished with q

$O(|V| + |E|) = O(n + e)$

Predecessor: Nil 1 1 2
 Nodes: 1 2 4 5
 Distance from 1: 0 1 1 2

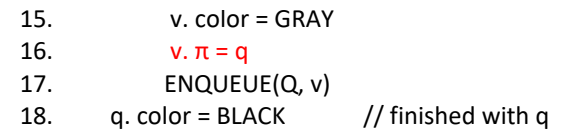
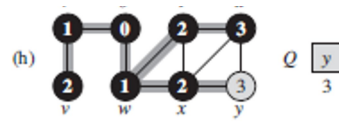


- Vertex with **WHITE** color means that it has not been visited yet.
- Vertex with **GRAY** color means that it has been visited but has not yet been fully explored.
- Vertex with **BLACK** color means that all of its adjacent vertices has also been visited (fully explored).



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$$\text{PRINT-PATH}(G; s; y) \text{ ----} \rightarrow \text{PRINT-PATH}(G; s; x) \text{ ----} \rightarrow \text{PRINT-PATH}(G; s; w) \text{ ----} \rightarrow \text{PRINT-PATH}(G; s; s)$$

y
x
w
s